

Dyadic-Comb Interpolation of the Fabius and Rvachev Functions

Runge instability, endpoint-flat Hermite constraints, exact cardinal obstructions,
and a logarithmically thin stabilization window

OpenAI GPT-5.6 Pro

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Abstract

Let $N = 2^n$ and interpolate the Fabius function F at the complete dyadic comb $x_j = j/N$, $0 \leq j \leq N$. Despite $F \in C^\infty$ and despite its flatness at both endpoints, the ordinary global Lagrange interpolant exhibits an unmistakable Runge transition: on an exact dyadic test grid its maximum error is about 3.80×10^{-3} at $N = 16$, 4.99 at $N = 32$, 3.59×10^7 at $N = 64$, and 3.84×10^{22} at $N = 128$. The oscillation is concentrated in narrow boundary layers. Exact rational computation of the endpoint derivative of the interpolant gives still stronger evidence: $\log_2 |P'_N(0)|$ is 949.99 at $N = 1024$ and 8070.57 at $N = 8192$, although $F'(0) = 0$.

We then impose the natural endpoint jets $P^{(r)}(0) = P^{(r)}(1) = 0$ for $1 \leq r \leq m$, while preserving every dyadic value. The resulting confluent interpolation problem has an exact factorization

$$P_{N,m}(x) = I_x(m+1, m+1) + [x(1-x)]^{m+1} Q_{N,m}(x),$$

where $Q_{N,m}$ is an ordinary interpolant on the interior comb. This converts the large confluent system into a rational barycentric problem and applies verbatim to the centered Rvachev bump, with the beta-polynomial bridge omitted. At $N = 64$ the Fabius error falls from 3.59×10^7 to 1.78×10^{-6} near $m = 14$, but rises again to 1.13×10^3 by $m = 24$. Matching first derivatives at every node is much worse: its error is already 1.54×10^7 at $N = 32$.

The reversal is explained by two exact cardinal modes. One mode carries boundary perturbations toward the center and decreases with m ; the other carries near-boundary perturbations toward a point just off the center and grows with m . Balancing them gives an explicit predicted order

$$m_{\text{bal}} + 1 = \frac{\log(A_N/B_N)}{\log(b_N/a_N)} \sim \frac{N \log 2}{\log N},$$

which predicts the experimentally optimal jet order surprisingly well. The admissible window has width only $\asymp N/(\log N)^2$. We prove, however, that no choice of $m = m_N$ stabilizes global equispaced interpolation in the worst-case sense: the weighted Lebesgue constant remains at least $\exp(cN)$ for all sufficiently large dyadic N . Thus endpoint flatness gives spectacular, structure-dependent finite-resolution cancellation, not a cure for the equispaced polynomial instability.

The report includes exact formulas, proofs, high-precision experiments, unproved conjectures on the Fabius-specific boundary layer and endpoint derivative defect, stable replacement algorithms, and a Lean formalization program. The accompanying Python source reproduces every numerical table and figure using exact rational dyadic samples.

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1 Provenance, audit boundary, and status discipline

1.1 Repository audit

The source boundary is the recursive documentation tree `Analysis/FabiusFunction/docs` in the public PROVE-IT repository. GitHub’s indexed snapshot at commit `fa53516af8cbe7d792a97b25378b228b96e37cf9` contained 84 TeX paths under this tree. The three commits between that index and the current main snapshot `3c7f2912a47433264f52ad530cc75adbaa83b764` changed no interpolation source and no TeX path relevant to the present question; the only modified TeX file records a new prefix-kernel counting formalization.

The audit distinguished living syntheses from frozen or duplicated historical revisions. The following were treated as the primary project state:

- `Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex`;
- `FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex`;
- `fabius_lean_walkthrough/fabius_lean_walkthrough.tex`;
- `non-formalized-research-frontiers/non-formalized-research-frontiers.tex`;
- the consolidated frontier drafts under `non-formalized-research-frontiers/drafts`;
- the archived dyadic, q -series, inverse-function, asymptotic, and representation studies, together with the transcribed source papers under `papers`.

Every TeX path was inventoried. Living expositions and every section found by the terms *Lagrange*, *interpolation*, *dyadic*, *Newton*, *Hermite*, *Runge*, *Richardson*, and *sampling* were read in detail; archived revisions were compared through their provenance and coverage ledgers rather than counted as independent prior results.

1.2 Novelty boundary

The audited corpus already contains exact dyadic arithmetic, Gaussian q -binomial interpolation on *geometric* nodes, finite-difference and Newton-basis formulas, Richardson transforms, all-orders endpoint asymptotics, Lambert- W coordinates, and Thue–Morse/Prouhet cancellation. It does *not* contain an analysis of global polynomial interpolation on the complete *uniform dyadic comb*, a Runge study for F or up, endpoint-flat confluent interpolation, the two cardinal obstructions derived below, or the universal instability theorem of [theorem 6.6](#). A search for “Runge” in the repository returned no result.

Claim labels. A statement marked **proved here** has a proof in this report. A numerical statement marked **numerically verified** is reproduced by the accompanying exact-rational/high-precision program. Statements marked **conjectural/open** are deliberately presented as conjectures or research problems, even when the numerical evidence is strong.

2 Executive findings

The answer to “does Runge’s phenomenon occur?” is yes in every operational sense tested. Ordinary global interpolation is initially accurate, then acquires enormous alternating boundary spikes. The evidence is stronger than a floating-point plot: all function values are exact rationals; increasing arithmetic precision does not change the result; and an exact rational endpoint derivative grows almost like 2^N .

The answer to “can higher derivatives suppress it?” has three layers.

1. Fixing the *natural flat endpoint derivatives* can suppress the actual Fabius or Rvachev error by many orders of magnitude at a given N .
2. The jet order must be tuned. Too few constraints leave the ordinary boundary mode; too many create a reverse mode from the center toward the boundary. The useful order is $m \asymp N/\log N$, in a window of width only $\asymp N/(\log N)^2$.
3. No tuning stabilizes arbitrary value data. The global operator remains exponentially ill-conditioned. Matching derivatives at every node is particularly destructive.

Main frontier result. For every sufficiently large dyadic N and every endpoint jet order $m \geq 0$, the weighted Lebesgue constant of the endpoint-flat value interpolation operator satisfies

$$\Lambda_{N,m} \geq Ce^{cN}$$

with absolute constants $c, C > 0$. Thus a growing endpoint jet can move and temporarily cancel the Runge instability for the special Fabius/Rvachev data, but cannot remove the underlying equispaced polynomial instability.

3 Fabius and Rvachev prerequisites

3.1 Normalization

We work on $[0, 1]$ with the standard Fabius function F , normalized by

$$F(0) = 0, \quad F(1) = 1, \quad F(1 - x) = 1 - F(x), \quad (3.1)$$

and satisfying

$$F'(x) = 2F(2x), \quad 0 \leq x \leq \frac{1}{2}. \quad (3.2)$$

The centered Rvachev bump is

$$u(x) := \text{up}(2x - 1) = F(1 - |2x - 1|) = F(2 \min(x, 1 - x)), \quad 0 \leq x \leq 1. \quad (3.3)$$

Then $F'(x) = 2u(x)$ on $[0, 1]$. Both F and u are C^∞ and flat at the relevant boundary points:

$$F^{(r)}(0) = F^{(r)}(1) = 0 \quad (r \geq 1), \quad u^{(r)}(0) = u^{(r)}(1) = 0 \quad (r \geq 0). \quad (3.4)$$

The same differential self-similarity that creates flatness also produces enormous global derivative norms. The audited project proves the estimate

$$\|F^{(r)}\|_\infty \leq 2^{r(r+1)/2}. \quad (3.5)$$

This tension—endpoint flatness versus superexponential derivative growth—is central to why elementary interpolation remainder estimates are misleading here.

3.2 Exact dyadic values

Write

$$V_n := F(2^{-n}), \quad V_0 = 1. \quad (3.6)$$

The exact inverse-power recurrence used in the experiments is

$$V_n = \frac{2^{-n(n-1)/2}}{2^n - 1} \sum_{k=0}^{n-1} \frac{2^{k(k-1)/2} V_k}{(n - k + 1)!}. \quad (3.7)$$

For $0 < a < 2^e$, set

$$b = \lfloor \log_2 a \rfloor, \quad r = e - b, \quad q = a - 2^b, \quad y = \frac{q}{2^e}, \quad (3.8)$$

and define

$$T_r(y) = \sum_{j=0}^r \frac{2^{j(j+1)/2}}{j!} V_{r-j} y^j. \quad (3.9)$$

The bit recursion is

$$F\left(\frac{a}{2^e}\right) = T_r\left(\frac{q}{2^e}\right) - F\left(\frac{q}{2^e}\right). \quad (3.10)$$

Equations (3.7)–(3.10) evaluate every sample in exact rational arithmetic. No numerical ODE, truncated Fourier product, or spline surrogate is used in the reported error scans.

4 Ordinary Lagrange interpolation on the dyadic comb

Let $N = 2^n$, $x_j = j/N$, and let $L_N f$ denote the unique polynomial of degree at most N with

$$(L_N f)(x_j) = f(x_j), \quad 0 \leq j \leq N. \quad (4.1)$$

4.1 Barycentric and Newton forms

For equispaced nodes the barycentric weights are proportional to $(-1)^j \binom{N}{j}$. Hence

$$(L_N f)(x) = \frac{\sum_{j=0}^N \frac{(-1)^j \binom{N}{j}}{x - j/N} f(j/N)}{\sum_{j=0}^N \frac{(-1)^j \binom{N}{j}}{x - j/N}}, \quad (4.2)$$

away from the nodes, with the usual exact-node convention. In forward-difference form,

$$(L_N f)(x) = \sum_{k=0}^N \Delta_{1/N}^k f(0) \binom{Nx}{k}. \quad (4.3)$$

The appearance of the ordinary binomial coefficients in (4.2), and the half-base q -binomial arithmetic in the dyadic samples themselves, already hints at a mixed $q = 1/q = 1/2$ transform; we return to this in [section 8](#).

4.2 Symmetry and an exact degree drop

Proposition 4.1 (Symmetry of the Fabius interpolant). *For even N ,*

$$(L_N F)(1 - x) = 1 - (L_N F)(x). \quad (4.4)$$

Consequently $L_N F$ has degree at most $N - 1$.

Proof. The polynomial $1 - (L_N F)(1 - x)$ takes the value $F(j/N)$ at every node by (3.1); uniqueness gives (4.4). Centering at $x = 1/2$ shows that $(L_N F)(1/2 + y) - 1/2$ is odd in y . Its a priori degree bound N is even, so the leading coefficient vanishes. $\square$

Equivalently, the top forward difference satisfies

$$\Delta_{1/N}^N F(0) = 0 \quad (N \text{ even}). \quad (4.5)$$

This cancellation is exact but does not protect the lower coefficients from exponential growth.

4.3 Endpoint derivatives and signed Stirling transforms

Let $s(k, r)$ denote the signed Stirling number of the first kind, $z^k = \sum_{r=0}^k s(k, r)z^r$. Differentiating (4.3) gives

$$(L_N f)^{(r)}(0) = r! N^r \sum_{k=r}^N \frac{s(k, r)}{k!} \Delta_{1/N}^k f(0). \quad (4.6)$$

For $f = F$ and even N , the $k = N$ term vanishes by (4.5). In particular,

$$(L_N F)'(0) = N \sum_{j=1}^N (-1)^{j+1} \binom{N}{j} \frac{F(j/N)}{j}. \quad (4.7)$$

Every term of (4.7) is rational, so the derivative defect can be computed exactly before taking a logarithm.

4.4 Observed Runge transition

The maximum errors below were sampled on the complete grid of mesh 2^{-12} , except the $N = 128$ pilot scan, which used mesh 2^{-11} . Every target value was exact.

Table 1: Ordinary global Lagrange interpolation of F on the complete dyadic comb.

N	degree bound	sampled $\ L_N F - F\ _\infty$	location of largest error
4	4	3.9966199×10^{-2}	boundary layer
8	8	9.4867660×10^{-3}	boundary layer
16	16	3.7966209×10^{-3}	boundary layer
32	32	4.9893489	$x \approx 0.00708$
64	64	3.5888912×10^7	$x \approx 0.00317$
128	128	3.8367529×10^{22}	$x \approx 0.00146$

This is the classical geometry of Runge instability: the central part remains visually reasonable while narrow endpoint oscillations explode. The special smoothness of F does not change the node geometry; indeed, because F is nonanalytic everywhere, the standard analytic-ellipse convergence mechanism is unavailable.

5 Endpoint-flat Hermite interpolation

5.1 The constrained problem

For $m \geq 0$, seek a polynomial $P_{N,m}$ satisfying

$$P_{N,m}(j/N) = F(j/N), \quad 0 \leq j \leq N, \quad (5.1)$$

$$P_{N,m}^{(r)}(0) = P_{N,m}^{(r)}(1) = 0, \quad 1 \leq r \leq m. \quad (5.2)$$

There are $N + 2m + 1$ scalar conditions, so the natural degree bound is $N + 2m$. A direct confluent Vandermonde solve would be both expensive and ill-conditioned. The flatness conditions instead permit an exact factorization.

Define the beta smoothstep

$$S_m(x) := I_x(m+1, m+1) = \frac{(2m+1)!}{(m!)^2} \int_0^x t^m (1-t)^m dt. \quad (5.3)$$

It is a polynomial of degree $2m + 1$ with

$$S_m(0) = 0, \quad S_m(1) = 1, \quad S_m^{(r)}(0) = S_m^{(r)}(1) = 0 \quad (1 \leq r \leq m), \quad (5.4)$$

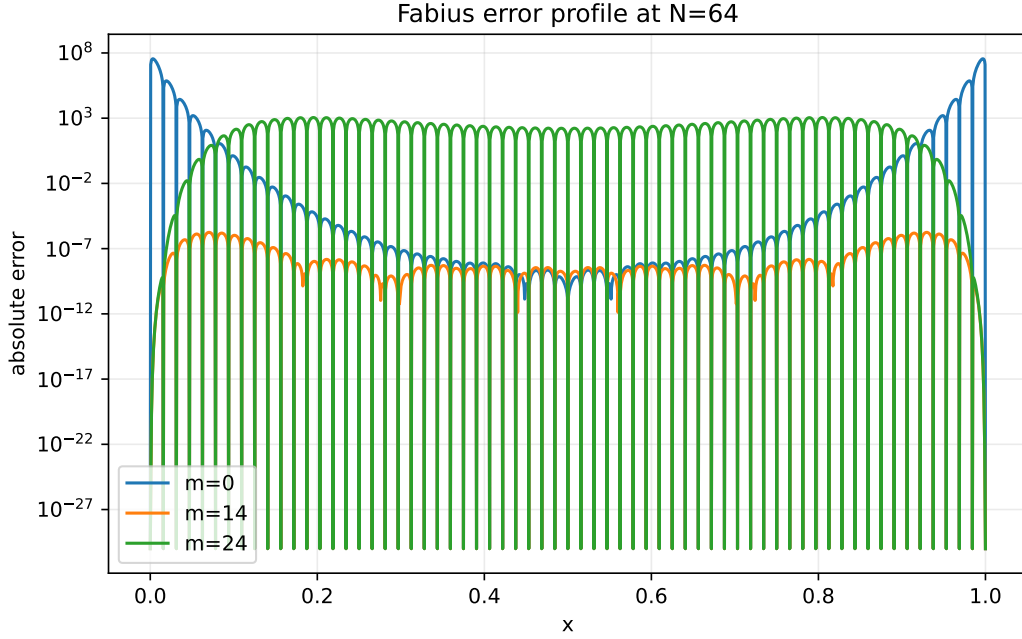


Figure 1: Absolute error at $N = 64$. The ordinary interpolant ($m = 0$) has a gigantic boundary spike; a tuned endpoint jet ($m = 14$) suppresses it; overconstraining ($m = 24$) creates a reverse instability.

and $S_m(1 - x) = 1 - S_m(x)$. Put

$$W_m(x) := [x(1 - x)]^{m+1}. \quad (5.5)$$

Theorem 5.1 (Exact endpoint-flat factorization). *There is a unique polynomial $P_{N,m}$ of degree at most $N + 2m$ satisfying (5.1)–(5.2). It is*

$$P_{N,m}(x) = S_m(x) + W_m(x)Q_{N,m}(x), \quad (5.6)$$

where $Q_{N,m}$ is the unique polynomial of degree at most $N - 2$ interpolating the interior data

$$Q_{N,m}(j/N) = \frac{F(j/N) - S_m(j/N)}{[\frac{j}{N}(1 - \frac{j}{N})]^{m+1}}, \quad 1 \leq j \leq N - 1. \quad (5.7)$$

For even N , $P_{N,m}(1 - x) = 1 - P_{N,m}(x)$ and its actual degree is at most $N + 2m - 1$.

Proof. If P satisfies the endpoint values and jets, then $P - S_m$ has zeros of multiplicity $m + 1$ at both 0 and 1. Hence $P - S_m = W_m Q$ for a polynomial Q . The degree bound on P implies $\deg Q \leq N - 2$, and the interior value conditions force (5.7); ordinary interpolation at $N - 1$ distinct points gives uniqueness. Conversely, (5.6) has all required values and jets. Symmetry follows by applying uniqueness to $1 - P_{N,m}(1 - x)$. Centered oddness removes the top even degree. $\square$

The beta polynomial can be evaluated without special functions:

$$S_m(x) = \frac{(2m + 1)!}{(m!)^2} \sum_{r=0}^m (-1)^r \binom{m}{r} \frac{x^{m+r+1}}{m + r + 1}. \quad (5.8)$$

Thus, when N is dyadic, all nodal quotient data in (5.7) are rational.

5.2 Rvachev version

For $u(x) = \text{up}(2x - 1)$, both endpoint values are zero. The bridge vanishes and the unique endpoint-flat interpolant is simply

$$U_{N,m}(x) = W_m(x)R_{N,m}(x), \quad (5.9)$$

where $R_{N,m}$ interpolates

$$R_{N,m}(j/N) = \frac{u(j/N)}{[\frac{j}{N}(1 - \frac{j}{N})]^{m+1}}. \quad (5.10)$$

Since $u(1 - x) = u(x)$, the quotient and interpolant are symmetric about $1/2$.

5.3 Barycentric implementation

For the interior nodes j/N , $1 \leq j \leq N - 1$, barycentric weights are proportional to

$$\tilde{w}_j = (-1)^{j-1} \binom{N-2}{j-1}. \quad (5.11)$$

Writing g_j for the quotient data, one evaluates

$$Q_{N,m}(x) = \frac{\sum_{j=1}^{N-1} \frac{\tilde{w}_j g_j}{x - j/N}}{\sum_{j=1}^{N-1} \frac{\tilde{w}_j}{x - j/N}}. \quad (5.12)$$

No confluent matrix is formed. This algorithm is exact at nodes, uses exact rational samples, and requires only arbitrary-precision arithmetic in the off-node evaluation.

5.4 Confluent remainder and why it is not predictive

Let

$$M = N + 2m + 1. \quad (5.13)$$

The classical Hermite remainder gives, for each x and some $\xi_x \in (0, 1)$,

$$F(x) - P_{N,m}(x) = \frac{F^{(M)}(\xi_x)}{M!} x^{m+1} (x-1)^{m+1} \prod_{j=1}^{N-1} \left(x - \frac{j}{N}\right). \quad (5.14)$$

Combining this with (3.5) produces a formally valid but enormous upper bound containing $2^{M(M+1)/2}$. It cannot explain the observed cancellation. The right diagnostic is not a derivative supremum but the data-amplification geometry developed next.

6 Weighted Lebesgue geometry and two exact obstructions

6.1 The relevant operator norm

Perturb only the $N - 1$ interior values, keeping endpoint values and endpoint jets exact. The perturbation of (5.6) is

$$\delta P(x) = W_m(x) \sum_{j=1}^{N-1} \frac{\delta f_j}{W_m(j/N)} \ell_j^{\text{int}}(x), \quad (6.1)$$

where ℓ_j^{int} is the ordinary cardinal polynomial for the interior comb. The corresponding pointwise and global Lebesgue functions are

$$\Lambda_{N,m}(x) = \sum_{j=1}^{N-1} \left| \frac{W_m(x)}{W_m(j/N)} \ell_j^{\text{int}}(x) \right|, \quad (6.2)$$

$$\Lambda_{N,m} = \max_{x \in [0,1]} \Lambda_{N,m}(x). \quad (6.3)$$

This is a lower bound on the conditioning of any implementation of the same polynomial operator; it is independent of the barycentric evaluation strategy.

6.2 A gamma formula for uniform cardinal functions

Let $y = Nx$. For $x \in (0, 1)$ not a node,

$$|\ell_j^{\text{int}}(x)| = \frac{\Gamma(y)\Gamma(N-y)|\sin \pi y|}{\pi|y-j|\Gamma(j)\Gamma(N-j)}, \quad 1 \leq j \leq N-1. \quad (6.4)$$

Indeed, reflection gives

$$\left| \prod_{k=1}^{N-1} (y-k) \right| = \frac{\Gamma(y)\Gamma(N-y)|\sin \pi y|}{\pi}, \quad (6.5)$$

and differentiation at $y = j$ gives $\prod_{k \neq j} |j-k| = \Gamma(j)\Gamma(N-j)$. Formula (6.4) makes the two competing modes completely explicit.

6.3 Boundary-to-center mode

Take $x = 1/(2N)$ and the central node $j = N/2$.

Proposition 6.1 (First exact cardinal obstruction). *For even N ,*

$$\Lambda_{N,m} \geq A_{N,m} := A_N a_N^{m+1}, \quad (6.6)$$

where

$$A_N = \frac{\Gamma(N - \frac{1}{2})}{\sqrt{\pi} (\frac{N}{2} - \frac{1}{2}) \Gamma(N/2)^2}, \quad a_N = \frac{2N-1}{N^2}. \quad (6.7)$$

Moreover,

$$A_N \sim \frac{2^N}{\pi \sqrt{2} N}, \quad A_{N,m} \sim \frac{2^{N+m+1}}{\pi \sqrt{2} N^{m+2}} \quad (m \text{ fixed}). \quad (6.8)$$

Proof. Keep only the $j = N/2$ term in (6.2). Formula (6.4) at $y = 1/2$ gives A_N . The weight ratio is

$$\frac{x(1-x)}{(1/2)(1/2)} = \frac{2N-1}{N^2} = a_N.$$

Stirling's formula yields (6.8). □

For every fixed m , the first mode is exponential. Endpoint flatness of fixed order can delay the Runge transition but cannot change its asymptotic character.

6.4 Center-to-boundary mode

Now take $x = 1/2 + 1/(2N)$ and the near-boundary node $j = 1$.

Proposition 6.2 (Second exact cardinal obstruction). *For even N ,*

$$\Lambda_{N,m} \geq B_{N,m} := B_N b_N^{m+1}, \quad (6.9)$$

where

$$B_N = \frac{\Gamma(N/2 - \frac{1}{2})^2}{\pi \Gamma(N-1)}, \quad b_N = \frac{N+1}{4}. \quad (6.10)$$

Moreover,

$$B_N \sim 2^{2-N} \sqrt{\frac{2}{\pi N}}. \quad (6.11)$$

Proof. At $y = N/2 + 1/2$ and $j = 1$, (6.4) reduces to B_N because $\Gamma(N/2 + 1/2) = (N/2 - 1/2)\Gamma(N/2 - 1/2)$. The weight ratio is

$$\frac{(\frac{1}{2} + \frac{1}{2N})(\frac{1}{2} - \frac{1}{2N})}{(\frac{1}{N})(1 - \frac{1}{N})} = \frac{N+1}{4} = b_N.$$

Stirling gives (6.11). □

The second mode is tiny at $m = 0$ but grows as approximately $(N/4)^m$. It is the precise mechanism behind the overconstrained blow-up.

6.5 The logarithmically thin balance window

Equating the exact bounds gives a distinguished real jet order.

Definition 6.3 (Two-mode balance order). For $N > 4$, define

$$m_{\text{bal}}(N) + 1 = \frac{\log(A_N/B_N)}{\log(b_N/a_N)}. \quad (6.12)$$

Proposition 6.4 (Asymptotic balance law). *As $N \rightarrow \infty$ through powers of two,*

$$m_{\text{bal}}(N) = \frac{N \log 2}{\log N} \left(1 + \mathcal{O}\left(\frac{1}{\log N}\right) \right). \quad (6.13)$$

The leading boundary-suppression and reverse-amplification thresholds are

$$m_-(N) + 1 \sim \frac{N \log 2}{\log(N/2)}, \quad (6.14)$$

$$m_+(N) + 1 \sim \frac{N \log 2}{\log(N/4)}, \quad (6.15)$$

and

$$m_+(N) - m_-(N) \sim \frac{N(\log 2)^2}{(\log N)^2}. \quad (6.16)$$

Proof. Insert (6.8) and (6.11) into (6.12). The numerator is $2N \log 2 + \mathcal{O}(\log N)$ and the denominator is $2 \log N + \mathcal{O}(1)$. Equations (6.14) and (6.15) follow by separately balancing the exponential parts of $A_{N,m}$ and $B_{N,m}$ against one. Their difference is an elementary expansion of the reciprocal logarithms. □

For a fixed available jet budget m , the leading inversion of $m \sim N \log 2 / \log N$ is

$$N_{\text{max}}(m) \sim -\frac{m}{\log 2} W_{-1}\left(-\frac{\log 2}{m}\right), \quad (6.17)$$

where W_{-1} is the lower real branch of Lambert's function. This is a direct bridge between the interpolation problem and the lower-branch Lambert- W machinery already central in the endpoint asymptotics of F .

Table 2: Exact two-mode balance order and the best tested jet order. The $N = 8$ Fabius minimum is a small-degree anomaly; from $N = 16$ onward the prediction is close.

N	m_{bal}	best tested m for F	best tested m for u
8	2.293	6	4
16	4.106	5	5
32	7.163	8	8
64	12.415	14	14
128	21.573	23 (selected scan)	—
256	37.758	—	—

6.6 No endpoint-jet sequence stabilizes the operator

The two special modes explain practical tuning, but by themselves their balanced common value can become small. A third, entropy-scale cardinal mode proves a stronger result. Let

$$H(\theta) = -\theta \log \theta - (1 - \theta) \log(1 - \theta) \quad (6.18)$$

be the binary entropy in natural logarithms.

Lemma 6.5 (Interior entropy growth). *Let N be divisible by 4, take $x_N = 1/4 + 1/(2N)$ and $j = N/2$. Then*

$$\log \left| \ell_{N/2}^{\text{int}}(x_N) \right| = N(\log 2 - H(1/4)) + \mathcal{O}(\log N). \quad (6.19)$$

Furthermore,

$$\frac{x_N(1 - x_N)}{1/4} \rightarrow \frac{3}{4}. \quad (6.20)$$

Proof. Apply Stirling's formula to (6.4). The exponential part of the gamma quotient is the difference between the entropy rates at $j/N = 1/2$ and $x_N \rightarrow 1/4$. The weight ratio is immediate. $\square$

Theorem 6.6 (Universal endpoint-flat instability). *There exist absolute constants $c, C > 0$ and N_0 such that, for every dyadic $N \geq N_0$ and every integer $m \geq 0$,*

$$\Lambda_{N,m} \geq C e^{cN}. \quad (6.21)$$

In particular, no sequence $m = m_N$ makes the endpoint-flat global interpolation operators subexponentially conditioned on the complete dyadic comb.

Proof. Set $\alpha = \log 2 - H(1/4) > 0$. Choose a sufficiently small constant $\varepsilon > 0$.

If $m \leq \varepsilon N$, keep the $j = N/2$ term at x_N from [theorem 6.5](#). Its logarithm is

$$N\alpha + m \log(3/4 + o(1)) + \mathcal{O}(\log N),$$

which is at least $c_1 N$ by the choice of ε .

If $m > \varepsilon N$, use the exact lower bound (6.9). By (6.11),

$$\log B_{N,m} \geq -N \log 2 + \varepsilon N \log(N/4) + \mathcal{O}(\log N),$$

which exceeds $c_2 N$ for all sufficiently large N (indeed it is eventually of order $N \log N$). Taking $c = \min(c_1, c_2)$ and absorbing finitely many prefactors into C proves the claim. $\square$

Corollary 6.7. *For each fixed m , the endpoint-flat interpolation operator is exponentially unstable; for every linear growth law $m \geq \varepsilon N$, it is eventually superexponentially unstable in N .*

This theorem is compatible with the spectacular actual errors in [section 7](#): a large operator norm need not be realized by the highly correlated, self-similar Fabius samples. It does mean that data noise, rounding, or a generic smooth perturbation can destroy the cancellation.

7 Numerical experiments

7.1 Protocol

The accompanying `numerical_experiments.py` implements (3.7)–(3.10) with Python’s exact `Fraction`. Interpolants are evaluated by scaled barycentric formulas in `mpmath`, generally at 120 to 300 decimal digits. Precision was increased until every displayed digit stabilized. Error maxima are sampled on all dyadic points of a stated fine grid, not on a random subset. The program also verifies symmetry, nodal reproduction, beta-bridge endpoint values, and several known small dyadic values before running the experiments.

The experiments distinguish two notions:

- *actual error*, obtained from the exact Fabius or Rvachev data;
- *worst-case amplification*, measured by the weighted Lebesgue function.

Conflating these notions would hide the most interesting result.

7.2 Fabius endpoint jets

Table 3: Ordinary and best tested endpoint-flat interpolation errors for F on a grid of mesh 2^{-12} .

N	ordinary error	best tested m	best tested error	improvement factor
8	9.48677×10^{-3}	6	2.09066×10^{-4}	4.54×10^1
16	3.79662×10^{-3}	5	2.73408×10^{-5}	1.39×10^2
32	4.98935	8	7.62257×10^{-7}	6.55×10^6
64	3.58889×10^7	14	1.77661×10^{-6}	2.02×10^{13}

At $N = 64$, selected values make the reversal explicit:

m	0	8	10	12	14	16	20	24
$\ P_{64,m} - F\ _\infty$	3.59×10^7	4.76×10^{-3}	1.56×10^{-4}	8.30×10^{-6}	1.78×10^{-6}	7.80×10^{-5}	4.73	1.13×10^3

(7.1)

The selected $N = 128$ scan gave errors 1.57×10^{-1} , 3.36×10^{-2} , 7.88×10^{-3} , 3.98×10^{-3} , and 1.70×10^{-2} at $m = 20, 21, 22, 23, 24$, respectively, against an ordinary error 3.84×10^{22} .

7.3 Rvachev endpoint jets

Table 4: Ordinary and best tested endpoint-flat interpolation errors for $u(x) = \text{up}(2x - 1)$.

N	ordinary error	best tested m	best tested error	improvement factor
8	2.02170×10^{-1}	4	1.33916×10^{-2}	1.51×10^1
16	1.96934×10^{-1}	5	6.50226×10^{-4}	3.03×10^2
32	4.49719×10^2	8	4.08484×10^{-5}	1.10×10^7
64	1.04564×10^{10}	14	6.19248×10^{-4}	1.69×10^{13}

The same optimal orders for $N = 16, 32, 64$ are notable because u uses no beta bridge. This supports the interpretation that the order is controlled primarily by node geometry and endpoint multiplicity, while the detailed target determines how completely the large modes cancel.

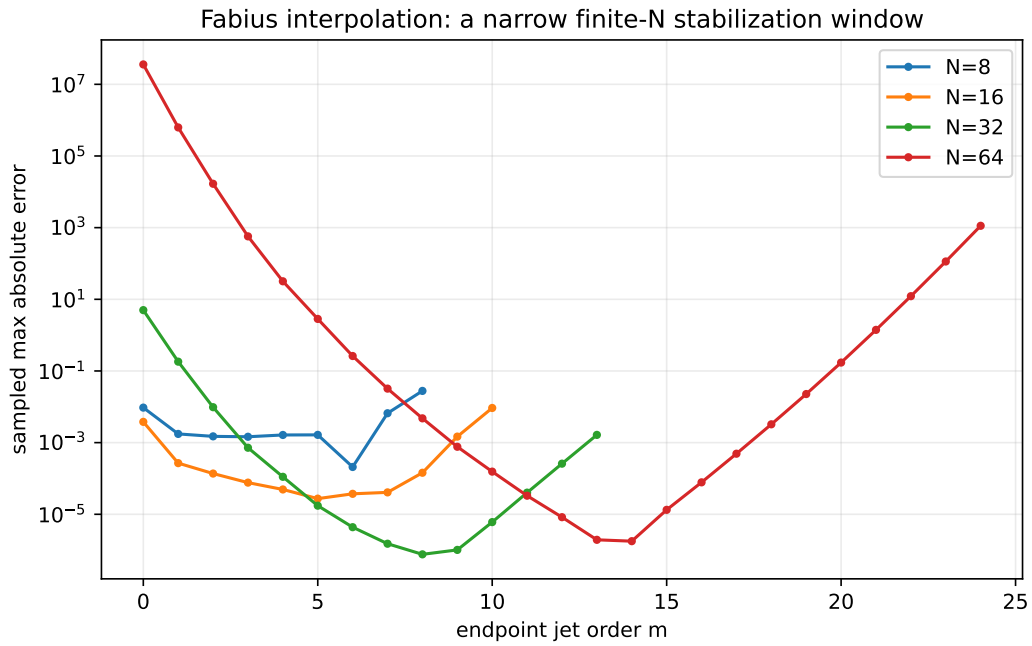


Figure 2: Fabius error versus endpoint jet order. Every sufficiently resolved curve is U-shaped: ordinary Runge growth is first suppressed, then a reverse mode appears.

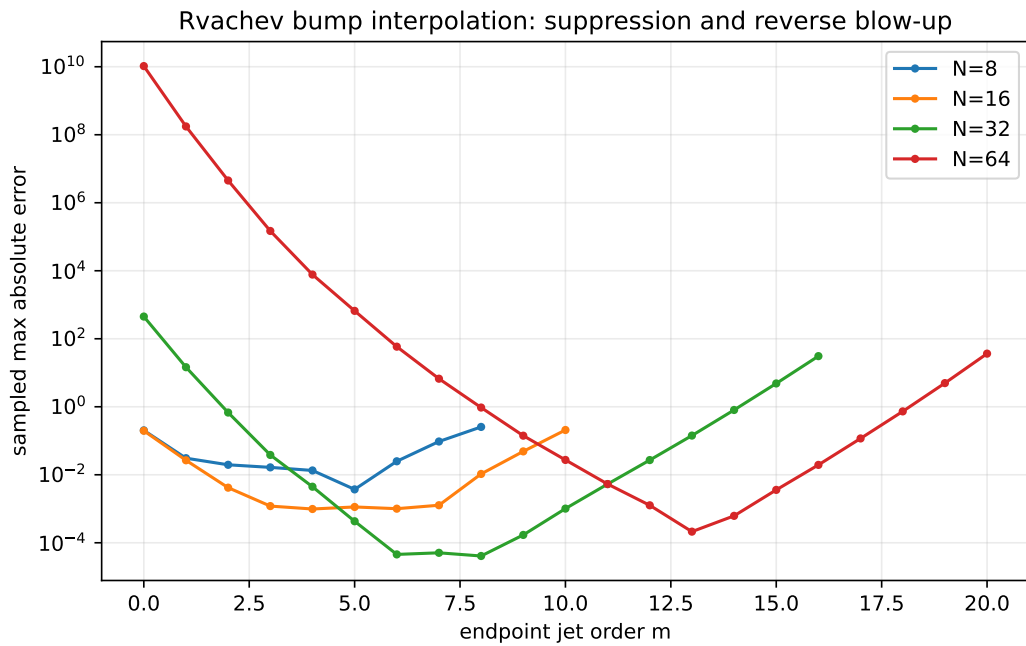


Figure 3: Endpoint-flat interpolation of the centered Rvachev bump.

7.4 Actual accuracy versus operator conditioning

At $N = 64$, a mesh- 2^{-11} scan of (6.3) gives

m	0	8	10	12	14	16	20	24
$\Lambda_{64,m}$	4.40×10^{16}	6.40×10^6	2.20×10^5	1.29×10^4	2.83×10^3	4.37×10^4	8.46×10^7	5.49×10^{11} .

(7.2)

The practical Fabius optimum $m = 14$ therefore still has a worst-case amplification of about 2825. The target's self-similar residual data happen to avoid the dangerous sign pattern.

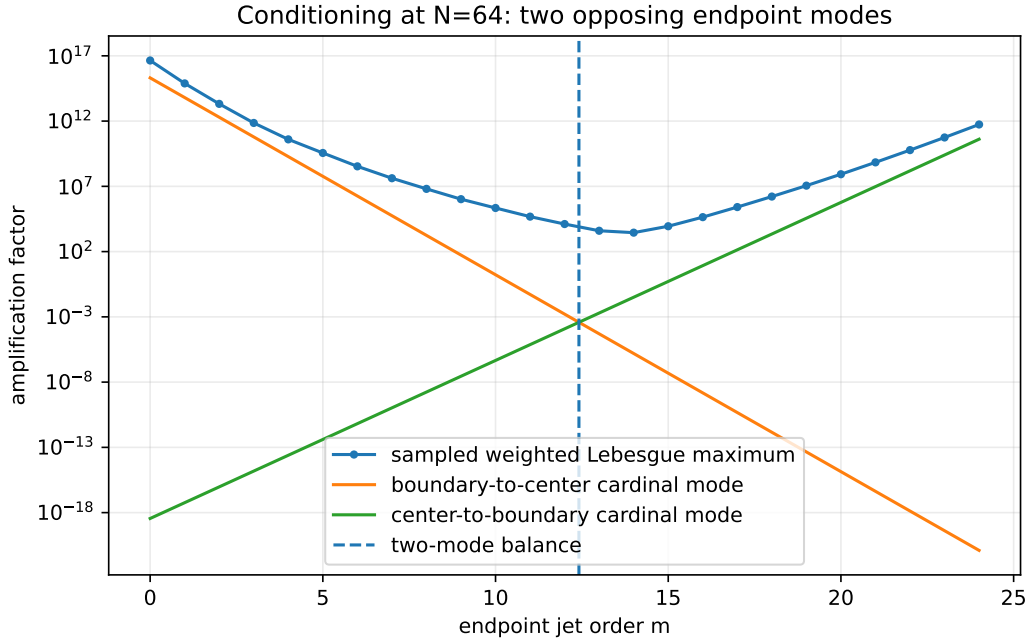


Figure 4: The sampled weighted Lebesgue maximum and the two exact cardinal lower modes at $N = 64$. Their crossing predicts the useful finite- N window.

7.5 Why fixing derivatives at every node is worse

One might instead match both $F(x_j)$ and $F'(x_j)$ at all $N + 1$ nodes, producing the standard first-order Hermite polynomial of degree at most $2N + 1$. With ordinary cardinal functions ℓ_j it is

$$H_N(x) = \sum_{j=0}^N [(1 - 2\ell'_j(x_j)(x - x_j))\ell_j(x)^2 F(x_j) + (x - x_j)\ell_j(x)^2 F'(x_j)]. \quad (7.3)$$

The dyadic derivatives are exact because

$$F'(j/N) = 2F(2 \min(j/N, 1 - j/N)). \quad (7.4)$$

The squared cardinal functions in (7.3) effectively square the already dangerous equispaced geometry, while the extra degree provides more room for oscillation. Higher derivative information is therefore not intrinsically stabilizing; *where* it is imposed matters more than the fact that it is exact.

7.6 Exact endpoint derivative defect

Equation (4.7) was evaluated as a reduced rational number. The following values show the scale; the sign is irregular.

Table 5: First-derivative Hermite interpolation at every dyadic node.

N	degree bound	sampled maximum error
4	9	1.14814×10^{-2}
8	17	3.73583×10^{-3}
16	33	2.27198
32	65	1.53809×10^7
64	129	1.35218×10^{23}

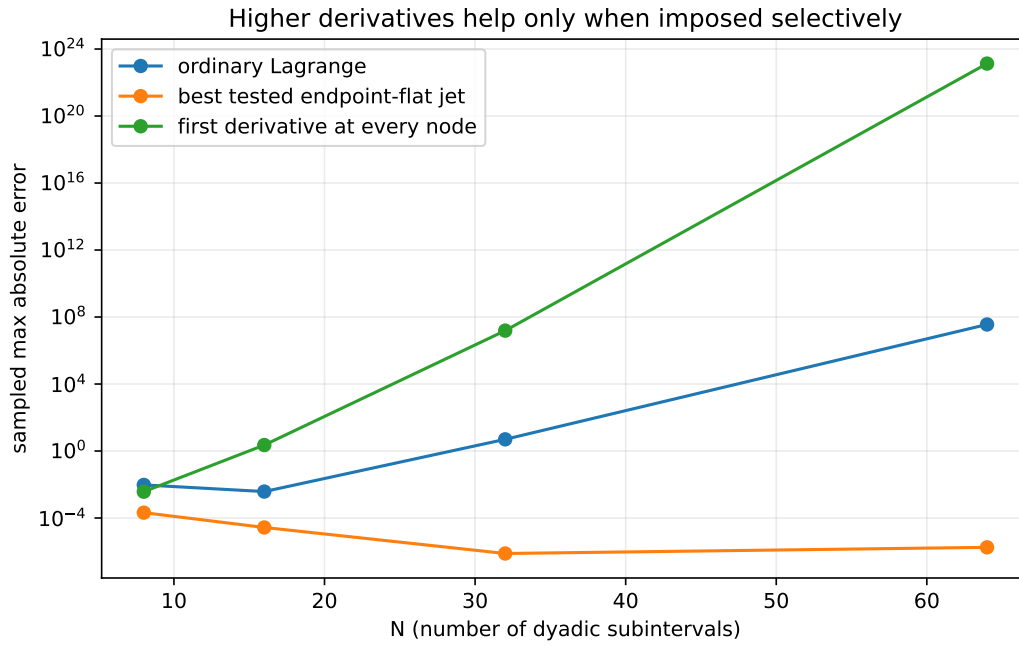


Figure 5: Ordinary Lagrange, best tested endpoint-flat interpolation, and all-node first Hermite interpolation.

Table 6: Exact endpoint derivative defect of the ordinary Fabius interpolant.

N	sign	$\log_2 (L_N F)'(0) $	$N - \log_2 (L_N F)'(0) $
16	+	-0.681	16.681
32	-	10.816	21.184
64	+	34.823	29.177
128	-	85.942	42.058
256	-	206.287	49.713
512	-	450.491	61.509
1024	+	949.989	74.011
2048	+	1957.083	90.917
4096	-	3992.945	103.055
8192	-	8070.574	121.426

The defect is a particularly clean observable because the true derivative is exactly zero. It is also a binomial transform of exact dyadic values, making it a natural meeting point of uniform-node interpolation and the project's Thue–Morse/ q -binomial arithmetic.

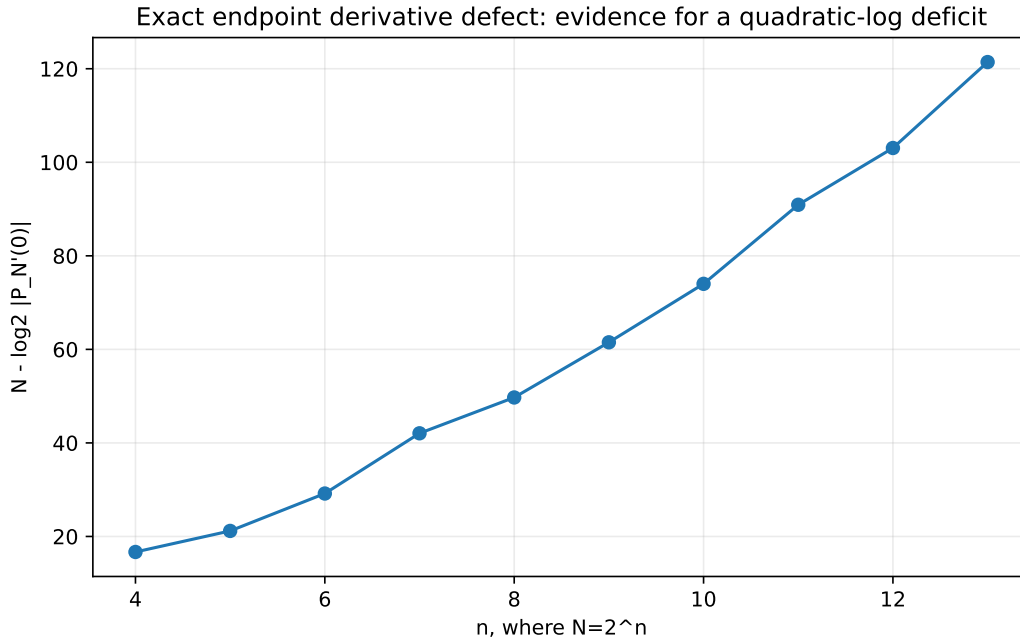


Figure 6: The binary deficit $N - \log_2 |(L_N F)'(0)|$. The data are consistent with a quadratic function of $\log_2 N$ plus lower-order oscillation.

8 Connections with the project's q -series and Thue–Morse structures

8.1 A mixed binomial transform

At uniform nodes, the interpolation weights are the ordinary Pascal row $(-1)^j \binom{N}{j}$ —a $q = 1$ object. The dyadic Fabius values are governed by $q = 1/2$ Pochhammer products and Gaussian binomial coefficients in the audited corpus. The endpoint defect

$$D_N = N \sum_{j=1}^N (-1)^{j+1} \binom{N}{j} \frac{F(j/N)}{j} \quad (8.1)$$

is therefore a mixed $q = 1/q = 1/2$ transform. It deserves study as an arithmetic object in its own right.

Writing $1/j = \int_0^1 t^{j-1} dt$ gives

$$\frac{D_N}{N} = \int_0^1 \sum_{j=1}^N (-1)^{j+1} \binom{N}{j} F(j/N) t^{j-1} dt. \quad (8.2)$$

This representation may permit a saddle analysis after inserting an exact dyadic-value formula. The alternating Pascal row has the same finite-difference cancellation that underlies Prouhet identities, but the nonpolynomial Fabius samples leave a small, highly structured residue.

8.2 Newton coefficients and Prouhet cancellation

The coefficients $\Delta^k F(0)$ in (4.3) annihilate polynomial trends of degree below k . For $N = 2^n$, the dyadic sample vector itself has recursive blocks controlled by binary digit patterns. It is plausible that the observed sign changes and the sublinear deficit in table 6 arise from a competition between:

- the entropy concentration of the Pascal row near $j = N/2$;
- exact Prouhet-type cancellation among dyadic blocks;
- the lower-Lambert endpoint scale governing $F(j/N)$ for small j ;
- a residual log-periodic phase inherited from the endpoint asymptotics.

A rigorous asymptotic for (8.1) would connect several currently separate parts of the project.

8.3 Bell and Bernoulli polynomials in refined balance laws

The leading balance law (6.13) uses only first-order Stirling expansions. All higher corrections to $\log A_N$, $\log B_N$, $\log a_N$, and $\log b_N$ are explicit Bernoulli-polynomial series. Reversion of (6.12) can be organized with complete Bell polynomials, exactly as in the project's all-orders Lambert- W endpoint expansions. Thus one can obtain

$$m_{\text{bal}}(N) \sim \frac{N \log 2}{\log N} \left(1 + \frac{c_1}{\log N} + \frac{c_2 (\log \log N)}{(\log N)^2} + \dots \right), \quad (8.3)$$

with computable coefficients. The exact formula (6.12) is already easy to evaluate, so the principal value of (8.3) is conceptual and formal: it places the interpolation window in the same asymptotic calculus as the Fabius endpoint.

9 What has and has not been proved about Runge divergence

The operator instability is proved in theorem 6.6. That theorem says there exist value perturbations whose interpolants grow exponentially, for every choice of endpoint jet order. It does not by itself prove that the *particular deterministic sequence* $L_N F$ diverges in supremum norm as $N \rightarrow \infty$.

For F , the numerical evidence is nevertheless overwhelming:

- direct exact-sample errors grow from 4.99 to 3.59×10^7 to 3.84×10^{22} as N goes from 32 to 64 to 128;
- the endpoint derivative defect is exact and nearly 2^N on a logarithmic scale;
- the spike location scales like $x \asymp 1/N$;
- increasing precision and refining the test grid leave the results unchanged.

A proof of the endpoint derivative conjecture below would immediately imply nonuniform convergence. With a quantitative Markov inequality, it would also force a large supremum norm of the interpolant in a boundary interval.

10 Conjectures and frontier problems

Conjecture 10.1 (Fabius Runge divergence). *For $N = 2^n$,*

$$\|L_N F - F\|_\infty \longrightarrow \infty. \quad (10.1)$$

More strongly, $\log_2 \|L_N F\|_\infty = N - o(N)$ along a subsequence.

Conjecture 10.2 (Endpoint derivative law). *There are positive constants c_1, c_2 such that*

$$N - c_2 (\log_2 N)^2 \leq \log_2 |(L_N F)'(0)| \leq N - c_1 (\log_2 N)^2 \quad (10.2)$$

for all sufficiently large dyadic N outside a possible sparse zero set. After subtracting the principal quadratic-log term, the remainder contains a bounded or slowly growing log-periodic component related to the endpoint phase of F .

The sign data suggest that a clean one-term asymptotic may require phase locking in $n = \log_2 N$, analogous to the phase-locked Richardson procedures already present in the repository.

Conjecture 10.3 (Optimal actual jet order). *Let $m_F(N)$ and $m_u(N)$ minimize the true supremum errors of $P_{N,m}$ and $U_{N,m}$. Then*

$$m_F(N) = m_{\text{bal}}(N) + \mathcal{O}\left(\frac{\log N}{\log \log N}\right), \quad m_u(N) = m_{\text{bal}}(N) + \mathcal{O}\left(\frac{\log N}{\log \log N}\right). \quad (10.3)$$

A stronger and numerically plausible version replaces the error term by $\mathcal{O}(1)$ along a phase-locked dyadic subsequence.

Conjecture 10.4 (Boundary-layer profile). *There exist normalizations A_N , B_N and a subsequence of dyadic N such that the ordinary interpolation error has a nontrivial scaled limit*

$$A_N((L_N F)(\xi/N) - F(\xi/N)) \longrightarrow \mathcal{E}(\xi) \quad (10.4)$$

locally for $\xi > 0$. The profile $\mathcal{E}$ is governed jointly by the entropy saddle of the uniform cardinal functions and the lower-Lambert endpoint asymptotic of F .

Open problem 10.5 (Arithmetic of the mixed transform). *Determine a closed form, recurrence, 2-adic valuation law, or Mahler equation for D_{2^n} in (8.1). Even a sharp upper and lower bound of the form (10.2) would settle the most basic Fabius-specific Runge question.*

Open problem 10.6 (Noise threshold). *For tuned $m = m_{\text{bal}}(N)$, quantify the largest relative perturbation of the exact dyadic samples that preserves the observed Fabius accuracy. The crude threshold $\Lambda_{N,m}^{-1}$ is pessimistic; the relevant structured condition number should project noise onto the self-similar residual subspace.*

11 Stable replacements for a global equispaced polynomial

The negative operator theorem does not make the dyadic comb unusable. It says only that using all samples in one high-degree polynomial is the wrong global representation. Several replacements preserve the exact dyadic arithmetic.

11.1 Piecewise dyadic Hermite splines

Divide $[0, 1]$ into dyadic blocks and use low-degree Hermite pieces. Derivatives at dyadic nodes are again exact Fabius values at a finer scale. One can enforce C^r continuity and retain strict locality. Because the polynomial degree stays fixed while the block size shrinks, there is no global Runge mechanism. This is the most direct practical method.

11.2 Constrained least squares with degree below N

The impossibility theorem of Platte, Trefethen, and Kuijlaars shows that stable exponential convergence from equispaced samples is unavailable, but stable root-exponential convergence is compatible with a polynomial degree of order $\sqrt{N}$. For Fabius data, solve a weighted least-squares problem of degree $d \asymp \sqrt{N}$ while imposing exact endpoint flatness. The beta-factorization still applies, but Q is fitted rather than interpolated. Oversampling converts the dangerous square system into a stable rectangular one.

11.3 Mapped Chebyshev and fake-node methods

Map the dyadic samples to approximate Chebyshev locations, or construct a polynomial in a nonlinear coordinate whose pullback nodes are near-Chebyshev. The map should respect $x \leftrightarrow 1 - x$ and the boundary flatness. A particularly natural family uses a Lambert coordinate fitted to the known endpoint scale, rather than a generic Kosloff–Tal–Ezer map.

11.4 Rational and Floater–Hormann interpolation

Local-degree rational barycentric formulas on equispaced nodes have much smaller Lebesgue constants and no polynomial endpoint catastrophe. Endpoint derivative constraints can be incorporated by rational Hermite formulas or by applying the rational interpolant to the quotient in (5.6). Poles must be monitored, but the method is a better candidate than global all-node Hermite interpolation.

11.5 Self-similar multiresolution bases

The infinite sinc product and Rvachev refinement equation suggest a basis adapted to the function rather than to generic algebraic polynomials. A multiresolution expansion in scaled translates of ψ would represent the dyadic recursion exactly and turn interpolation into a sparse refinement problem. This direction is closest to the intrinsic geometry of the target.

11.6 Inverse Fabius interpolation

Direct polynomial interpolation of F^{-1} near 0 and 1 is even more delicate because the inverse has singular derivatives. The lower-Lambert variable should be used first: interpolate the slowly varying correction after removing the known asymptotic inverse. The present endpoint-flat polynomial is not the right tool for the inverse problem.

12 Lean formalization program

The core new results are well suited to formalization because they separate algebraic, finite, and asymptotic layers.

1. **Algebraic factorization.** Define the beta smoothstep as the finite polynomial (5.8); prove endpoint jets, symmetry, divisibility by $x^{m+1}(1-x)^{m+1}$, and uniqueness of [theorem 5.1](#).
2. **Uniform cardinal formula.** First prove finite-product versions of the two special cardinal evaluations. The gamma formulas can be added later; the exact products already suffice for the lower bounds.
3. **Two obstruction theorems.** Formalize [theorems 6.1](#) and [6.2](#) with factorials and central binomial coefficients. Translate to gamma notation as a corollary.
4. **Universal instability.** Use explicit Stirling bounds for factorials rather than asymptotic notation. The dichotomy $m \leq \varepsilon N$ versus $m > \varepsilon N$ is elementary once the finite cardinal products are available.
5. **Exact computational certificates.** Import selected rational error values for small N as kernel-checked examples, not as theorems about continuum maxima.
6. **Endpoint defect.** Prove (4.7) from the barycentric basis and connect it to the existing finite-difference and dyadic-value modules.

A useful module decomposition would be

Module	Contents
DyadicCombLagrange	nodes, barycentric weights, symmetry, Newton form
EndpointFlatHermite	beta bridge, factorization, uniqueness
WeightedLebesgue	perturbation operator and cardinal functions
DyadicCombObstructions	exact A and B modes
DyadicCombInstability	exponential lower-bound theorem
FabiusInterpolationDefect	exact endpoint derivative transform

13 Reproducibility and archive contents

The accompanying archive contains:

- this LaTeX source and the compiled PDF;
- `numerical_experiments.py`, with detailed comments and command-line tasks;
- `requirements.txt`;
- all CSV outputs under `data`;
- all vector figures under `figures`;
- `README.md` with exact reproduction commands;
- SHA256SUMS for integrity checking.

The principal commands are:

```
python numerical_experiments.py --task smoke
python numerical_experiments.py --task fabius --N 64 --m-start 0 --m-end 24 \
  --grid-level 12 --output data/fabius_N64.csv
python numerical_experiments.py --task up --N 64 --m-start 0 --m-end 20 \
  --grid-level 12 --output data/up_N64.csv
python numerical_experiments.py --task lebesgue --N 64 --m-start 0 --m-end 24 \
  --grid-level 11 --output data/lebesgue_N64.csv
python numerical_experiments.py --task figures \
  --data-dir data --figure-dir figures
```

14 Conclusions

Global equispaced interpolation of the Fabius and Rvachev functions is a particularly clean laboratory for Runge instability. The target is infinitely smooth, its dyadic samples are exact, its endpoints are flat to every order, and yet ordinary interpolation blows up. This removes several common numerical excuses and exposes the node geometry directly.

Endpoint-flat Hermite constraints do something genuinely remarkable: at $N = 64$ they turn a 10^7 error into a 10^{-6} error. But they do so only in a narrow order window. The exact factorization by $[x(1-x)]^{m+1}$ reveals two opposing cardinal modes; their balance predicts $m \asymp N/\log N$ and explains both the suppression and the reversal. A separate entropy mode proves that no order sequence stabilizes arbitrary data.

The most promising theoretical frontier is now sharply posed: explain why the special Fabius residual vector cancels an exponentially ill-conditioned operator so efficiently, and derive the endpoint derivative transform asymptotically from the existing Thue–Morse, q -binomial, and Lambert- W machinery. The most promising practical route is not a still higher global polynomial, but a local or oversampled representation adapted to the dyadic self-similarity.

A Derivation of the interior cardinal formula

Let

$$\omega_N(y) = \prod_{k=1}^{N-1} (y - k). \quad (\text{A.1})$$

For noninteger $y \in (0, N)$,

$$\Gamma(y)\Gamma(N-y) = \Gamma(y)\Gamma(1-y) \prod_{k=1}^{N-1} (k-y) \quad (\text{A.2})$$

$$= \frac{\pi}{\sin \pi y} (-1)^{N-1} \omega_N(y). \quad (\text{A.3})$$

Taking absolute values gives the numerator in (6.4). At an integer j ,

$$|\omega'_N(j)| = (j-1)!(N-j-1)! = \Gamma(j)\Gamma(N-j). \quad (\text{A.4})$$

Since

$$\ell_j^{\text{int}}(x) = \frac{\omega_N(Nx)}{(Nx-j)\omega'_N(j)}, \quad (\text{A.5})$$

formula (6.4) follows.

B Asymptotics of the two exact modes

Stirling's expansion in logarithmic form is

$$\log \Gamma(z) = \left(z - \frac{1}{2}\right) \log z - z + \frac{1}{2} \log(2\pi) + \sum_{r=1}^{R-1} \frac{B_{2r}}{2r(2r-1)z^{2r-1}} + \mathcal{O}(z^{-2R+1}). \quad (\text{B.1})$$

Substitution into (6.7) gives

$$\log A_N = N \log 2 - \log N - \log(\pi\sqrt{2}) + \mathcal{O}(N^{-1}), \quad (\text{B.2})$$

and substitution into (6.10) gives

$$\log B_N = -(N-2) \log 2 + \frac{1}{2} \log \frac{2}{\pi N} + \mathcal{O}(N^{-1}). \quad (\text{B.3})$$

Also

$$\log a_N = -\log(N/2) + \log(1 - 1/(2N)), \quad (\text{B.4})$$

$$\log b_N = \log(N/4) + \log(1 + 1/N). \quad (\text{B.5})$$

These four expansions yield all assertions in [theorem 6.4](#). Retaining further Bernoulli terms and reverting the quotient with Bell polynomials yields the all-orders program described in [section 8](#).

C Selected raw error data

Table 7: Selected Fabius endpoint-flat errors. The full table is in `data/fabius_endpoint_flat.csv`.

N	m	degree bound	sampled max error
16	0	16	3.7966209×10^{-3}
16	3	22	7.6539608×10^{-5}
16	5	26	2.7340776×10^{-5}
16	8	32	1.4436091×10^{-4}
32	0	32	4.9893489

N	m	degree bound	sampled max error
32	4	40	1.1112922×10^{-4}
32	6	44	4.3679563×10^{-6}
32	8	48	7.6225669×10^{-7}
32	10	52	6.0546854×10^{-6}
32	13	58	1.6440974×10^{-3}
64	0	64	3.5888912×10^7
64	8	80	4.7635400×10^{-3}
64	10	84	1.5556810×10^{-4}
64	12	88	8.3033837×10^{-6}
64	14	92	1.7766068×10^{-6}
64	16	96	7.7955030×10^{-5}
64	18	100	3.2453804×10^{-3}
64	20	104	4.7281168
64	24	112	1.1263086×10^3

D Corpus crosswalk

The present report relies on the following established project components but does not rederive them:

Corpus component	Role here
Fabius_Function_and_Rvachev_Up.tex	Normalization, exact dyadic values, relation between F and up, flatness, derivative growth, Fourier/infinite-sinc background.
FabiusFunction_Mathematical_Glossary.tex	Notation and cross-references for q -Pochhammer, Gaussian binomial, Bell/Bernoulli, Lambert- W , and Thue-Morse structures.
fabius_lean_walkthrough.tex	Existing formal theorem inventory and suggested module dependencies.
non-formalized-research-frontiers.tex	Prior frontier claims, especially geometric-node interpolation, Richardson methods, endpoint asymptotics, and spectral representations.
archive/research-frontiers/Fabius_Dyadic_q_Connections/...	Half-base q -series arithmetic used to frame the mixed transform.
archive/research-frontiers/Fabius_Dyadic_Formulae_to_Asymptotics/...	Dyadic-value and endpoint-asymptotic bridge.
papers/arXiv-1702.06487v3/157-Arithmetic-v3.tex	External arithmetic background on dyadic Fabius values.

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